

which automatically switches between local and remote sensing depending on whether the sense leads are correctly connected.

e-noses and e-tongues, as their names suggest, are inspired by the neurophysiology of smell and taste, and attempt to mimic the abilities of their human counterparts. These technologies automate evaluation of samples with complex composition and are able to recognise specific properties and characteristics.

In animals, sensory information is processed by the neural system. Likewise, data collected through selective sensor arrays must be analysed by pattern recognition tools that employ various mathematical and statistical processing techniques. Such systems can provide quantitative results and, in some cases, are even able to detect differences that a human sensory panel cannot distinguish.

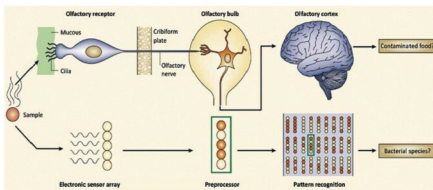
In food analysis, arrays of gas sensors are termed e-noses, while arrays of liquid sensors are referred to as e-tongues. The first scientific literature on these systems appeared in the 1980s but it has only been in the last decade, that special attention has been given to emerging technologies in electronic senses due to the food industry's progressive interest in rapid at- and on-line analysis of product quality and safety.

Typically, e-nose instruments use four types of sensors: conducting polymers, metal-oxide semiconductors (MOS), metal-oxide semiconducting field-effect transistors (MOSFETs) and oscillating sensors such as quartz-crystal microbalances.

e-tongue instruments generally use the following analytical solutions: mass sensors, which are miniaturised solid-state devices that exploit piezoelectric effect; potentiometric methods, for example, ion-selective electrodes; and voltammetric or optical sensors, in which an indicator molecule changes its optical properties when exposed to a target analyte. Hybrid e-tongues, based on a combination of potentiometry, voltammetry and conductivity, offer great potential.

Industrial applications

Electronic senses offer advantages of consistency as well as not having to use



e-nose devices mimic the human olfactory system (Courtesy: www.csi.cmu.edu)

humans for what may sometimes be unpleasant or even dangerous tasks.

Up to now, electronic senses have had straight industrial applications. For instance, food, beverage, pharmaceuticals, plastics, packaging and environmental industries have clear uses for electronic senses.

In food and beverage industry, e-noses can detect when flavours or odours in food are 'off,' and if there has been contamination or migration of compounds from plastic packaging into content. MOS odour sensors can also control how fast a flavour is diluted, ensure quality control or reverse-engineer an aroma. Electronic tongues can assess things like bitterness, taste stability over time or product adulteration.

Concrete applications of e-noses can be found in the analysis of meat flavours, volatile organic compounds (VOCs) formed during post-harvest ripening of fruits, ham product evolution during storage and olive oil defects, as well as identification of geographical origin of foodstuffs.

In the last decade, for example, Barilla has successfully implemented MOS-based e-noses in different quality control labs to recognise residual solvents and to continuously monitor the various plastic food-packaging materials adopted within bakery production sites. At Barilla, e-noses are, in fact, used as the first appraiser of packaging quality, which limits the number of gas chromatographic confirmatory analyses solely to the samples marked as uncertain or bad by the MOS instrument.

e-tongues can be used to monitor and discriminate among mineral water, coffee and soft drink samples. Reported applications of e-tongues in

food analysis cover process monitoring, foodstuff recognition/characterisation, evaluation of 'freshness,' quality control and authenticity assessments.

Electronic senses were applied to evaluate the effect of brewing temperature on sensorial properties of espresso coffees (ECs) produced by a bar machine that was able to work with constant, increasing and decreasing water temperature profiles. The ECs were analysed by e-nose, e-tongue and e-eye to depict their aroma and taste fingerprint, and to evaluate the visual characteristics of foam. Physico-chemical analyses were carried out to determine the extraction rate of typical EC components and evaluate their antioxidant activity.

Electronic devices coupled with multivariate statistical analysis demonstrated a good ability to discriminate and characterise coffee samples on the basis of their sensorial properties in relation to the brewing temperature. According to these results, electronic senses can be applied to assess the influence of percolation parameters on the sensory attributes of ECs, thus resulting in useful tools for optimisation of processing conditions.

Sensory characteristics of food products are gaining importance in the international food research and industry to meet consumer needs. Food producers make efforts to improve the quality and offer a wide range of foods. Comparison with existing brands in sensory point-of-view and developing products in sync with consumer needs are essential for the successful new product launch.

Usually a trained sensory panel is used to evaluate product quality. How-